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J. Chem. Phys. 164, 044107 (2026)

<https://doi.org/10.1063/5.0292167>



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Cite as: J. Chem. Phys. 164, 044107 (2026); doi: 10.1063/5.0292167

Submitted: 21 July 2025 • Accepted: 5 January 2026 •

Published Online: 23 January 2026



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Note: This paper is part of the JCP Special Topic, Michele Parrinello Festschrift.

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ABSTRACT

Machine-learning-based interatomic potentials are widely employed in atomistic simulations, but they struggle to capture long-range electrostatic correlations, which are ubiquitous in polar and biomolecular systems. We present a physics-informed machine-learning interatomic potential that incorporates long-range electrostatic interactions through a polarizable framework. Our model combines two equivariant message-passing neural networks: one for short-range interactions and the other for environment-dependent atomic dipoles. The model is trained not only on energies and forces but also on Born effective-charge tensors, enabling accurate predictions of field-induced properties such as infrared absorption spectra and LO–TO phonon splittings. We validate the method on ionic solids (NaCl), liquid water, and halide perovskites (MAPbI₃), demonstrating improved modeling of long-range polarization effects while maintaining competitive accuracy in energy and force predictions. Our results highlight the necessity of explicit long-range electrostatics for capturing collective phenomena in insulating and polar materials.

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I. INTRODUCTION

The development of accurate interatomic potentials, i.e., of models to calculate energies and atomic forces in molecular simulations, is at the core of efforts to improve the accuracy and the predictability of computational approaches to materials science, biology, chemistry, and Earth and planetary sciences. Reaching quantum-chemical accuracy in potentials for systems with a large number of atoms is an ongoing effort that started more than half a century ago, also thanks to the seminal study of Car and Parrinello, who introduced a clever and computationally efficient method to evolve the ground-state wave function during the atomic dynamics.¹ Despite these efforts, the computational load of *ab initio* force fields, i.e., of interatomic potentials where forces and energies are determined “on the fly” with quantum chemical or density-functional theory (DFT) accuracy,² limits the application of these methods to systems containing up to a few thousand atoms and for times well below a nanosecond, which are insufficient to tackle a vast range of interesting problems. To this aim, considerable efforts have been made in the last decades to develop *parameterized* force fields,

that is, potentials where energies and forces are obtained from a suitable, arbitrarily complex functional form, parameterized on the basis of experimental data and/or quantum chemical calculations. The shortsightedness of interatomic interactions has greatly helped in the development of parameterized potentials, in the early days by providing a physically sound guide to the choice of the functional forms, and more recently by informing the selection of local descriptors in machine-learning approaches.^{3–5} Machine-learning interatomic potentials (MLPs) are at present widely employed in molecular simulations and have revolutionized the field by achieving near-quantum-mechanical accuracy at a fraction of the cost of traditional *ab initio* methods. Using regression techniques⁴ or neural networks,³ MLP approximates the total potential-energy surface as a sum of atomic contributions,

$$E = \sum_i E_i^a(\mathcal{D}_i), \quad (1)$$

where each descriptor $\mathcal{D}_i$ encodes the local atomic environment of atom i . The local environment comprises all neighboring atoms j

within a cutoff sphere of radius r_{cut} centered on atom i . Descriptors typically include pairwise distances, angles formed by triplets, optional higher-order geometric terms, and the elemental identities of those neighbors. By restricting interactions to this finite region, MLPs maintain linear scaling with system size while accurately capturing the most chemically significant short-range interactions.

However, purely local descriptors struggle to capture long-range correlations, leading to inadequate treatment of electrostatic interactions between distant, non-neutral entities such as ions.⁶ Consequently, accurately modeling the structural and dynamical behavior of polar systems and biomolecular assemblies remains challenging for standard MLPs.^{7,8}

To address this limitation, various methods have been developed to equip short-range (SR) MLPs with nonlocal information. One prominent approach is the message-passing network,⁹ where atomic descriptors are refined through successive message-passing steps, gradually expanding each atom's effective receptive field. Although adding more layers should, in principle, allow the incorporation of information from increasingly distant neighbors,⁹ in practice, two issues limit network depth. First, over-squashing occurs when the growing volume of information from distant atoms must be compressed into fixed-size hidden representations, impeding the accurate transmission of critical long-range structural information.¹⁰ Second, over-smoothing emerges when repeated aggregation drives all descriptors toward a common value, erasing subtle structural distinctions and reducing model expressivity.¹¹ Together, these effects constrain the ability of message-passing interatomic potentials to faithfully represent true long-range interactions. An alternative approach is based on the consideration that nonlocal information need not be propagated through multiple message-passing layers but can instead be directly embedded into each atomic descriptor.^{6,12} This single-step enrichment equips each local representation with an awareness of distant environments, thereby avoiding the depth and complexity of deep message-passing architectures. Consequently, descriptor-level embeddings offer both computational efficiency and enhanced robustness in modeling long-range correlations. However, no stringent test has been performed yet to prove their ability to capture long-range electrostatic effects.

A completely different approach to include long-range interactions in MLP consists of explicitly embedding electrostatic physics via Ewald summation.¹³ In these “physics-inspired” schemes, the long-range effects of local charge rearrangements are included by introducing atom-centered charges that are either explicitly dependent on local descriptors¹⁴ or are obtained through a charge-equilibration scheme¹⁵ in which electronegativity and chemical hardness are expressed as neural networks as a function of local descriptors.^{7,8,16} From a computational perspective, both schemes present challenges; in the case of environment-dependent charges, the difficulty is enforcing the physical constraint of global charge neutrality, while in the case of charge equilibration, it is connected with the computational overload associated with the self-consistent procedure required to reach the equilibration.

In the above Ewald-based schemes, the atomic charges are treated as dynamical variables that vary along the trajectory and must be recalculated at every time step. While this is certainly a requirement in systems with a vanishing electronic excitation gap, where the response function of the electronic density remains finite at small $\mathbf{q}$ and the electronic charge redistribution takes place in

principle at all spatial scales, in systems with a finite excitation gap the charge-density response function vanishes for $\mathbf{q} \rightarrow 0$, and the electronic charge redistribution following the atomic displacements can, to a large degree, be captured by “local” rearrangements of the electronic charge. Zhang *et al.*,¹⁷ for example, have modeled the local redistribution of the electronic charge by decomposing the charge in terms of Wannier functions and introducing additional, fixed point charges located in the centroids of the Wannier functions, which remain in the vicinity of the reference atom or ion and whose positions are learned via an environment-dependent neural network. In the same spirit, a local, charge-neutral charge redistribution can be approximated, at long distances, by its dipolar term, which can be obtained by knowing the polarizability of each individual atom or ion. In fact, a vast literature describes attempts to develop *polarizable* potentials for water and other systems, parameterized on the basis of DFT or higher-level quantum chemical calculations.^{18–20} In these potentials, the atomic charge is fixed *a priori* to the nominal ionic charge or is obtained through a fit to a training dataset but is then kept constant during a simulation, and each species is associated with a polarizability and, therefore, with a time-dependent dipole moment that is typically obtained through a self-consistent procedure.

Here, we propose a new MLP “polarizable” model that is based on two distinct equivariant message-passing neural networks, one describing short-range interactions and one describing the polarization of each atom or ion. The total energy of the systems is expressed as a sum of the short-range term and of a long-range electrostatic term obtained from the (fixed) charges and the varying, “learned” dipoles. We expect the model to work reasonably well in all cases where previous non-machine-learned polarizable models have proven effective and more generally in all systems characterized by a non-vanishing electronic gap, where the electronic charge redistribution can be assumed to take place within the spatial range captured by local descriptors.

Our paper is organized as follows. In Sec. II, we introduce our computational approach, detail our training strategy, and describe implementation specifics, including dataset construction and hyperparameter choices. Section III benchmarks our method on NaCl using a classical reference potential and then extends the approach to liquid water and the halide perovskite MAPbI₃. Finally, in Sec. IV, we discuss our results and outline directions for future study.

II. COMPUTATIONAL METHOD

A. The equivariant message-passing neural network interatomic potential

We consider a periodic system of N atoms with positions $\{\mathbf{r}_i\}_{i=1}^N$. The total energy is split into short- and long-range contributions,

$$E_{\text{tot}} = E_{\text{short}} + E_{\text{long}}. \quad (2)$$

To model the short-range term E_{short} , we use an E(3)-equivariant message-passing neural network.⁹ Each atom i carries an initial embedding $\mathbf{h}_i^{(0)}$ encoding its element type. Neighbors within a cutoff r_{cut} are defined by

$$\mathcal{N}(i) = \{j \mid \|\mathbf{r}_j - \mathbf{r}_i\| < r_{\text{cut}}\}. \quad (3)$$

Over L layers, messages are computed and aggregated as

$$\mathbf{m}_i^{(l+1)} = \sum_{j \in \mathcal{N}(i)} \phi_{\text{int}}(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{r}_{ij}), \quad (4)$$

$$\mathbf{h}_i^{(l+1)} = \phi_{\text{upd}}(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l+1)}), \quad (5)$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. Choosing ϕ_{int} and ϕ_{upd} to be E(3)-equivariant ensures that $\mathbf{h}_i^{(l)}$ transforms covariantly under rotations and translations. After L layers, a readout neural network R maps $\{\mathbf{h}_i^{(L)}\}$ to per-atom energies,

$$E_{\text{short}} = \sum_{i=1}^N R(\mathbf{h}_i^{(L)}). \quad (6)$$

In this study, we adopt the NequIP architecture,²¹ which efficiently encodes both the scalar and geometric features in every layer.

The long-range energy E_{long} is computed by Ewald summation with explicit monopoles and dipoles.^{14,17} Each atom/ion is associated with a fixed charge q_i and an environment-dependent dipole. The atomic dipole is expressed as an equivariant message-passing neural network as follows:

$$\mathbf{p}_i = \sum_{j \in \mathcal{N}(i)} \phi_{\text{dipole}}(\mathbf{h}_i^{(L)}, \mathbf{h}_j^{(L)}, \mathbf{r}_{ij}). \quad (7)$$

By expressing the atomic dipole as a function of local descriptors, we assume that its value depends on the local environment of the atom. However, the message-passing nature of the neural network ensures that the spatial range captured by the model extends over several nearest-neighbor shells. We remark that the model developed in Zhang *et al.*¹⁷ is also based on a local dependence of the Wannier function centroids and, therefore, of the charge distribution. The long-range nature of the electrostatic interactions is captured in our model by the Ewald summation, where charges and dipoles interact at arbitrarily large distances.

To ensure that the model captures the relevant physics of long-range electrostatic interactions, the model is trained by minimizing a loss function that contains not only energy and atomic forces but also the atomic Born effective-charge (BEC) tensors, defined as

$$Z_{\kappa,\alpha\beta}^* = \frac{\partial P_{\alpha}^{\text{tot}}}{\partial u_{\kappa\beta}}, \quad (8)$$

where $u_{\kappa\beta}$ is the displacement of atom κ along Cartesian direction β , and the total polarization is given by

$$\mathbf{P}^{\text{tot}} = \sum_i (q_i \mathbf{r}_i + \mathbf{p}_i). \quad (9)$$

Constraining the model to reproduce the effective charges is essential not only to ensure the correct physical behavior of long-range forces but also to be able to extract from the model information about the response of the system to external electrical fields, including modeling the infrared (IR) absorption spectra, as we will see later.

B. Implementation and training

All calculations were carried out with an in-house reimplementation of the NequIP framework,²¹ augmented by an explicit

long-range electrostatics module. Short-range interactions are handled by an E(3)-equivariant message-passing network where interatomic distances r_{ij} are expanded in a Bessel radial basis and smoothly attenuated via a polynomial cutoff, while angular dependence is captured through its spherical harmonics up to order $l = 2$. The node features used in this study comprise $64 \times 0e + 64 \times 1o + 64 \times 2e$, while the maximum number of message-passing layers is fixed at 3. All message-passing and readout operations utilize the e3nn²² and PyTorch²³ libraries, and neighbor-list construction as well as molecular dynamics workflows are managed via the atomic simulation environment package.²⁴

Models were trained for up to 500 epochs with a batch size of one structure, reshuffling the training set at the start of each epoch. Optimization employed the AdamW algorithm²⁵ with an initial learning rate of 0.025, which was reduced by a factor of 0.8 whenever the validation loss failed to improve for ten consecutive epochs. To enhance convergence stability, we maintained an exponential moving average (EMA) of the model parameters with a decay coefficient of 0.99.

Because the energy, force, and Born-effective-charge losses differ substantially in scale, we applied a geometric-decay schedule to their respective weights. Denoting the current epoch by t and the total number of epochs by T , each weight $\lambda(t)$ follows:

$$\lambda(t) = \lambda_{\text{init}} \left(\frac{\lambda_{\text{final}}}{\lambda_{\text{init}}} \right)^{t/T}, \quad (10)$$

ensuring $\lambda(0) = \lambda_{\text{init}}$ and $\lambda(T) = \lambda_{\text{final}}$. We set all final weights to 1 and chose initial values $\lambda_E = 10^{-2}$, $\lambda_F = 10^3$, and $\lambda_Z = 10^{-2}$ for energy, force, and Born-effective-charge losses, respectively. This schedule guarantees that each loss component contributes appropriately during the early phase of training while converging to balanced importance in the final model.

III. RESULTS

A. Solid NaCl

We first validate our model on bulk NaCl, a paradigmatic ionic crystal whose longitudinal-transverse optical (LO-TO) phonon splitting at the Brillouin-zone center (Γ point) directly reflects long-range Coulomb interactions. In LO modes, out-of-phase cation-anion displacements generate a macroscopic electric field that stiffens the restoring force and raises the LO frequency, whereas TO modes, which produce no net longitudinal field in the long-wavelength limit, are unaffected by the non-analytic Coulomb term.

For the sake of simplicity, we train our MLP on a classical potential consisting of a Coulomb term with fixed point charges of $+0.85e$ (Na) and $-0.85e$ (Cl) and of a Lennard-Jones term.²⁶ To assess system-size transferability, we train the model exclusively on 64-atom NaCl supercells and validate its transferability to larger supercells. The Lennard-Jones parameters are listed in Table I.

The reference training set comprises uncorrelated snapshots drawn every 10 000 steps from molecular-dynamics (MD) trajectories at 300 K and ambient pressure (timestep = 2 fs), generated with LAMMPS.²⁷ Although training on larger cells would more directly capture long-range behavior, we restrict to 64-atom cells to probe the model's ability to extrapolate electrostatic physics to

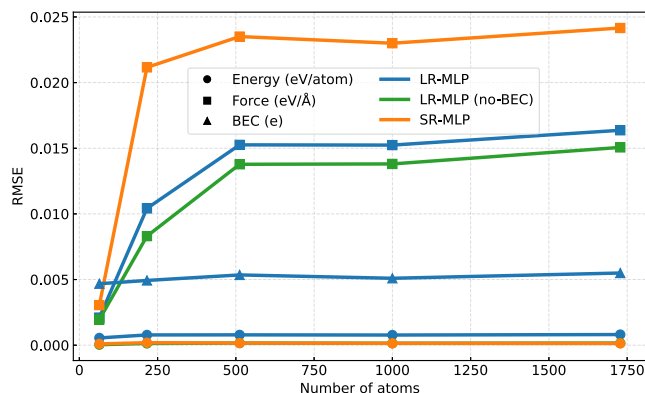
TABLE I. Lennard-Jones parameters for NaCl employed in this study are those of the Madrid-2019 force field,²⁶ with a cutoff radius fixed at 10 Å.

Pair	ϵ (eV)	σ (Å)
Na–Na	0.015 259 9	2.217 37
Cl–Cl	0.000 797 25	4.699 06
Na–Cl	0.014 913 0	3.005 12

much larger systems—a test rarely performed due to the prohibitive cost of large-scale DFT. In a prior study on bcc iron at pressures up to 600 GPa, we demonstrated that a model trained on 128 atoms faithfully reproduced properties in 1024-atom cells, reflecting the short-range nature of metallic bonding.²⁸ Here, we apply the same protocol to NaCl to determine whether an ML potential trained on small insulating cells can recover long-range Coulomb effects in extended crystals.

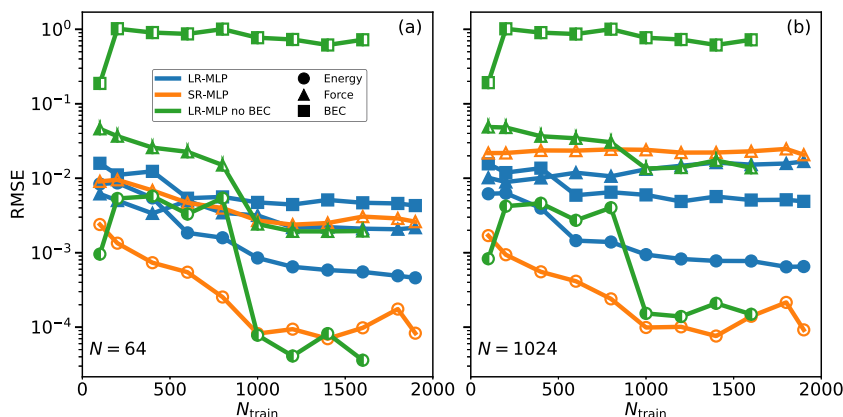
As shown in Fig. 1, 10^3 training configurations are largely sufficient to converge the root-mean-squared errors (RMSEs) of energies, forces, and BECs in the model with full long-range interactions (LR-MLP). In addition, Fig. 1 shows that the performance of the model is preserved when the size of the validation set increases to 1000 atoms. We confirmed the scalability of the model with system size by generating an independent validation set of NaCl configurations—ranging from 64 to 1728 atoms at 300 K and ambient pressure, using the same classical Lennard-Jones plus Coulomb potential and MD protocol. Figure 2 shows the performance of the LR-MLP for increasing supercell size. The RMSEs in energy and Born effective charges are small (<1 meV/atom and $5 \times 10^{-2}e$, respectively) and independent of size. In contrast, the force RMSE increases from about 2 meV/Å at 64 atoms (the training-cell size) to about 15 meV/Å around 500 atoms and then plateaus for larger systems.

In order to test the contribution of the long-range term in the model, we also considered a model based exclusively on the short-range term in Eq. (2). This short-range model (SR-MLP) was trained on the same set of data as the LR model. As shown in Figs. 1 and 2, the short-range model yields marginally lower energy errors but performs rather poorly in reproducing forces, whose error increases by 50% compared to the full long-range model.

**FIG. 2.** Root-mean-square errors for energy, force, and Born effective charge as a function of the size of the validation set for (i) the short-range (SR) equivariant message-passing potential (orange line) and (ii) the long-range (LR) potential with Ewald electrostatics, trained with Born effective charges (blue line) or without BEC (green line).

Finally, we assessed the impact of including the Born effective charge constraint on the model's ability to reproduce energies and forces. From an optimization perspective, adding a constraint to the loss function implies a degradation of the performance on the other terms (energy and forces), so it is important to check that the degradation is minimal. As shown in Fig. 2, a long-range model trained using only energy and force terms (without Born effective charges) performs only marginally better than the model trained with the additional constraint. We therefore conclude that enforcing this physics-based constraint does not compromise the model's accuracy on energies and forces. On the other hand, the LR model trained without BEC data yields totally incorrect values for the Born effective charges (see Fig. 1), indicating that effective charges cannot be accurately inferred from models trained exclusively on energies and forces.

Enforcing a physical constraint on effective charges leads to a substantial improvement in the description of the electric-field-induced excitations of the system. Figure 3 compares the phonon excitation frequencies of NaCl obtained with the three models

**FIG. 1.** Root-mean-squared errors (RMSEs) for energy (eV/atom), forces (eV/Å), and Born effective charge (BEC, |e|) as a function of the number of training configurations. The training configurations contain 64 atoms. RMSEs are calculated on two validation datasets containing 64 (left) and 1024 (right) atoms, respectively. Learning curves are shown for three models: the short-range (SR) and long-range (LR) machine-learning interatomic potentials (MLPs) and a LR-MLP trained without BEC data.

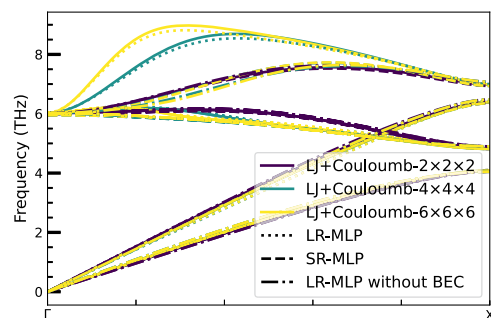


FIG. 3. LO-TO phonon splitting in NaCl computed for four interaction models: (i) classical Lennard-Jones with fixed point charges (solid line), which is our reference model; (ii) short-range (SR) machine-learning potential (MLP) (dashed line); (iii) long-range (LR) MLP trained with Born effective charges (BEC) (dotted line); and (iv) LR-MLP trained without BEC (dashed-dotted line).

described earlier. To mimic the typical situation of a phonon calculation performed with periodic boundary conditions (with finite displacements or with molecular dynamics), the non-analytic Coulomb contribution to the phonon frequencies at $q \rightarrow 0$ is not included. Both the long-range MLP and the classical reference potential exhibit a pronounced LO-TO splitting that extends to low wavelengths (and vanishes as $q \rightarrow 2\pi/L$, where L is the supercell size, due to periodic boundary conditions¹⁷). In contrast, the short-range MLP alone produces a spectrum that is invariant to cell size and lacks a clear LO-TO splitting at low wavelengths.

B. Liquid water

As an additional test of the ability of our physically constrained model to reproduce response functions, we used the method to generate a model for water and applied the model to compute the infrared absorption spectrum of liquid water. The model was trained on the publicly available dataset of Schmiedmayer and Kresse (Ref. 29). We selected the subset generated with the RPBE-D3 functional, which yields vibrational frequencies in closer agreement with experiment. The dataset comprises 100 uncorrelated snapshots sampled uniformly from an MD trajectory; we used 95 for training and 5 for validation.

Our long-range MLP achieved a training-set RMSE of 0.4 meV/atom for total energies and 25 meV/Å for forces,

matching the accuracy of both regression baselines and state-of-the-art message-passing models³⁰ (Fig. 4). The RMSE on the Born effective-charge tensor was $3.5 \times 10^{-2}e$, also comparable to previous studies.²⁹ By contrast, a pure short-range MLP trained on the same data attained the same energy RMSE (0.4 meV/atom) but a lower force error (about 20 meV/Å), in agreement with earlier observations that adding long-range terms does not always improve short-range accuracy in water.¹⁴ However, we note that we have not performed a learning-curve analysis to determine whether the long-range model requires a larger training set than the 95 configurations used in this study for training to reach convergence. This omission motivates more systematic studies of dataset size to fully quantify the benefits of including long-range electrostatics in aqueous systems.

To compute the infrared absorption spectrum, we performed molecular-dynamics simulations of 64 water molecules at 300 K in the canonical (NVT) ensemble. The system was equilibrated for 10 ps using a Nosé-Hoover thermostat³¹ with a coupling period of 100 fs. After equilibration, we collected a 100 ps production trajectory with a timestep of 0.25 fs in the microcanonical (NVE) ensemble, recording atomic positions, velocities, and instantaneous Born effective charges every 1 fs. We also verified energy conservation in the NVE ensemble: over a 1 ns simulation, the total energy drift remained below 2 meV per atom, confirming the numerical stability and reliability of our implementation.

We then applied the Green-Kubo relation for the IR absorption coefficient as given in Eq. (5),

$$\chi(\omega) = \frac{\beta}{3 V c \epsilon_0} \Re \int_0^T \langle \dot{\mathbf{P}}(0) \cdot \dot{\mathbf{P}}(t) \rangle e^{-i(\omega - i\delta)t} dt, \quad (11)$$

where $\beta = 1/(k_B T)$, V is the simulation cell volume, c is the speed of light, and ϵ_0 is the vacuum permittivity. The time derivative of the total polarization is computed on-the-fly via

$$\dot{\mathbf{P}}(t) = \sum_i \mathbf{Z}_i^*(t) \mathbf{v}_i(t), \quad (12)$$

with $\mathbf{Z}_i^*(t)$, the instantaneous Born effective-charge tensor for atom i , and $\mathbf{v}_i(t)$, its velocity. The brackets $\langle \cdot \rangle$ denote an ensemble average over multiple microcanonical segments.

Figure 5 shows the infrared absorption spectrum of liquid water, obtained by averaging three independent 100 ps MD trajectories. Before computing the Fourier transform, each total

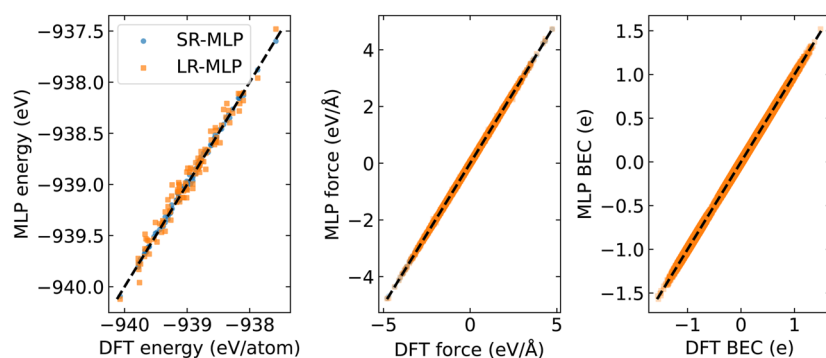


FIG. 4. Comparison of predictions from long-range (LR) and short-range (SR) machine-learning interatomic potentials (MLPs) with reference density functional theory (DFT) calculations²⁹ for water: (a) total energy per configuration, (b) force components (all three Cartesian components per atom), and (c) Born effective-charge tensor components (all nine tensor elements per atom). Each marker represents a single data point in the corresponding dataset.

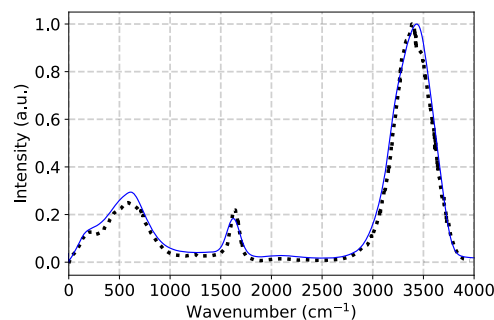


FIG. 5. Comparison of the infrared absorption spectrum of liquid water at ambient pressure and 300 K (full line, our study) with the one calculated in Ref. 29 (dashed line).

polarization-moment derivative autocorrelation function was multiplied by a Gaussian window with a decay constant of δ , as shown in Eq. (11), to improve peak resolution. Our spectrum captures the high-frequency O–H stretching band and the mid-frequency bending band with high fidelity and correctly reproduces the low-frequency librational and collective modes associated with intermolecular hydrogen-bond dynamics. These comparisons demonstrate both the statistical convergence of our approach and its quantitative agreement with established benchmarks. We stress that the results of Ref. 29 were obtained with a hybrid approach where a short-range MLP drives the dynamics and a second MLP predicts Born effective charges. Within our approach, the Born effective charges are extracted from the same model used to generate the energies and forces.

Finally, in order to demonstrate the effectiveness of our approach in capturing long-range screening effects in more complex geometries, we considered the liquid–vapor water interface.¹⁴ For this particular test, we refined the short-range term of our model using the training dataset for interfacial water by Niblett *et al.*³² We simulated interfacial water at 300 K following the setup described in Ref. 14. The slab system has dimensions of $24.8 \times 24.8 \times 120.0 \text{ \AA}^3$

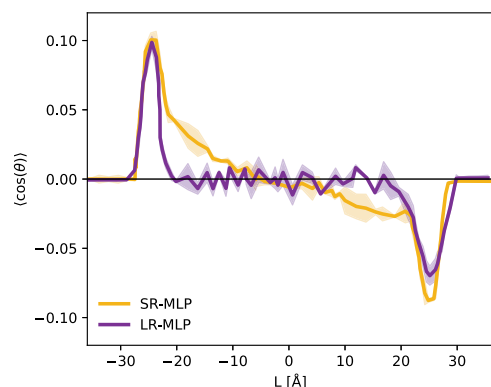


FIG. 6. Orientation of the water dipole moments at liquid–vapor water interfaces (measured as the cosine of the angle the dipole moment forms with the interface normal), as obtained with the short-range (SR) and the long-range (LR) model, respectively.

and contains 1024 water molecules. To minimize statistical noise, averages were obtained from five independent 500-ps-long simulations initialized with different atomic velocities. For comparison, we performed simulations with both the short-ranged (SR) and long-ranged (LR) versions of our model. We evaluated the orientational order profiles, shown in Fig. 6, by computing the cosine of the angle between the water dipole vectors and the z-axis (the longest direction in the simulation cell). At the interface, water molecules form a dipole layer, which is screened by the underlying layers, causing vanishing net dipole moments in the bulk. Without long-range interactions, this screening effect is not adequately captured: the SR model exhibits an extended dipole ordering that penetrates deep into the bulk. In contrast, the LR model recovers the physically correct orientational order profile, with a bulk region of almost vanishing net polarization, in agreement with the results of Ref. 14.

C. Halide perovskite

As a final validation, we apply our MLP to methylammonium lead iodide (MAPbI_3), which undergoes entropy-driven phase transitions from orthorhombic to tetragonal at 160 K and from tetragonal to cubic at 330 K. Including organic constituents (C, H, N) introduces additional chemical complexity—especially in the dynamic behavior of the methylammonium cation—providing a stringent test of our methodology.

Our training dataset comprises 300 DFT-computed structures (150 orthorhombic, 150 tetragonal). We randomly reshuffled and split these into 290 configurations for training and 10 for validation. On the training set, the ML potential achieves RMSEs of 0.3 meV atom^{−1} for energies and 14 meV Å^{−1} for forces. On the held-out validation set, the Born effective-charge tensor is predicted with an RMSE of 0.04 |e|, modestly outperforming previous benchmarks.²⁹ These results demonstrate the model's capability not only to capture both energetic and polarization properties across structurally distinct phases of MAPbI_3 but also to correctly represent the predominantly ionic character of the lead–iodide lattice while flexibly accounting for subtle molecular dipolar contributions.

We applied the same procedure as for liquid water to compute the infrared absorption spectrum of MAPbI_3 . The resulting spectrum shows good agreement with prior simulations,²⁹ although modest discrepancies in peak intensities and positions are observed

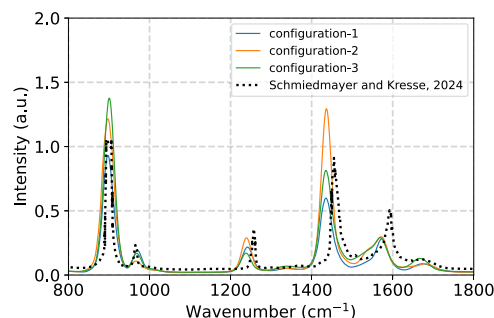


FIG. 7. Comparison of the infrared absorption spectrum of MAPbI_3 with the one calculated in Ref. 29. Configurations were extracted from short isobaric–isothermal simulations at ambient pressure and 107 K, in line with prior atomistic simulation studies,²⁹ for the orthorhombic phase.

(Fig. 7). In particular, the bands between 1200 and 1600 cm^{-1} are shifted to higher frequencies relative to previous results. While an insufficient training dataset may play a role—as noted for water—errors in the predicted Born effective charges (which directly scale absorption intensities) and slight inaccuracies in the MLP (which can shift or distort vibrational frequencies) could also contribute. Further investigation, including functional benchmarking and more extensive polarization training data, will be required to fully resolve these deviations.

IV. DISCUSSION AND CONCLUSIONS

We have presented a DFT-trained polarizable model for the interatomic potential of insulating solids based on a neural network approach to describe short-range interactions and induced dipoles. In two test cases, lead-halide perovskites and liquid water, incorporating explicit long-range electrostatics yield only marginal improvements in the accuracy with which the model reproduces DFT energies and forces, with respect to models based exclusively on short-range interactions. The lack of a clear improvement is possibly a consequence of the ability of the short-range term to capture interactions at medium-long distances through its message-passing layers. We find, however, that the explicit inclusion of long-range interactions makes the model appropriate to compute also the field-induced response of the system, as long as it is trained to reproduce also atomic effective charges. In general, we believe that forcing the long-range term of the model to acquire a physical significance, beyond the perfunctory role of improving forces and energies, is crucial whenever long-range electrostatics are indispensable for capturing collective aspects of the material behavior and response. In this paper, we presented three specific examples (the LO–TO splitting in an ionic solid, as well as the infrared absorption and the interfacial orientational order in water). However, one can think of several other situations where inclusion of long-range terms is crucial. In polar and ionic solids, non-analytic dipole–dipole couplings set phase-transition temperatures and dielectric constants—properties that short-range models alone cannot reproduce.³³ Likewise, in aqueous and ionic fluids, transport coefficients such as diffusion and conductivity emerge from correlated charge motions; omitting Coulomb interactions leads to significant discrepancies with experimental results.³⁴ Furthermore, lattice thermal conductivity in polar crystals hinges on LO–TO phonon splitting and anharmonic phonon lifetimes, both controlled by long-wavelength electrostatic fields.³⁵ Therefore, even when short-range fits achieve similar local force errors, only models that include explicit long-range electrostatics can faithfully reproduce the collective polarization fluctuations underlying macroscopic phase transitions, transport phenomena, and dielectric response.

ACKNOWLEDGMENTS

We acknowledge CINECA (Consorzio Interuniversitario per il Calcolo Automatico) under the ISCRA initiative and the Leonardo early access program for the computing resources and support. The research was supported by MONSTER, a project of the Italian National Centre for HPC, Big Data and Quantum Computing (ICSC) funded by the European Union—Next Generation EU, through PNNR.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Z. Li: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **S. Scandolo:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The datasets generated and analyzed during the current study, as well as the corresponding code, are available from the corresponding author upon reasonable request.

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